Science inquiry skills

Chapter
10

- plan and conduct investigations
- analyse data, draw conclusions, evaluate investigation design and findings
- evaluate the impact of advancements in human biology on individuals and society
- communicate understandings of human biology.

MULTIPLE-CHOICE QUESTIONS

1. *(2012:1.28)*

A lack of Vitamin D is said to be detrimental to the maintenance of bone density. If a scientist wanted to determine whether this view was supported by scientific evidence, which of the following would be the best method of doing so?

(a) carry out a survey of people with normal bone density to find out whether they take Vitamin D tablets or not

(b) compare the bone density of two groups of people with low bone density after providing only one of the groups with Vitamin D

(c) compare the incidence of people with low bone density in two different populations

(d) carry out an experiment in which a group of people with normal bone density are exposed to very little sunlight to prevent the production of Vitamin D

The next two questions refer to the information below.

For a number of years, there has been a belief among some people that the immunisation of infants and children causes autism. Many studies have focused on potential links between the ingredients in vaccines and their possible side effects. Mercury has been one of the ingredients tested. Numerous long-term experiments have been conducted to determine whether there is a link between autism and the presence of mercury in vaccines. However, no link has been established.

2. *(2015:1.21)*

Which of the following is a logical hypothesis for such an experiment?

(a) Autism is caused by mercury.

(b) Vaccines without mercury are better for infants and children.

(c) Vaccines containing mercury do not cause autism.

(d) Do vaccines containing mercury cause autism?

3. *(2015:1.22)*

The dependent variable in this experiment would be the

(a) volume of vaccine administered.

(b) number of infants and children in the study.

(c) amount of mercury in the vaccine.

(d) number of new cases of autism.

7. **[13 marks]** *(2013:2.36)*

A recent study investigated the effects of MDMA (ecstasy) on human thermoregulation. The research team selected 10 individuals (male and female) aged between 18 and 35 years. Each participant attended two sessions, one week apart. At the first session participants received, in tablet form, either a placebo or 2 mg/kg of MDMA. If at the first session participants received a placebo, at the second session they received MDMA, and vice versa.

At each session, participants assembled in a room at 10 am with the room temperature at 23°C. After 30 minutes, the room temperature was changed to 30°C, which took about 2 minutes. Data collection began at 11 am, when the drug or placebo tablet was given with a small amount of water. Recordings of core body temperature were taken every hour for the next 4 hours, with the session concluding at 3 pm.

Data for each of the **five** time periods were averaged and recorded (in order of time starting at 11 am) in a notebook as follows:

MDMA: 36.9°C, 37.1°C, 37.5°C, 37.6°C, 37.6°C
Placebo: 36.9°C, 37.0°C, 37.0°C, 37.1°C, 37.1°C

(a) Present the above data in a table. [5]

(b) Formulate a hypothesis for this experiment. [1]

8. **[10 marks]** *(2014:2.35)*

An experiment was conducted on the effects of fluid consumption on urine production. The experiment involved the comparison of water consumption with the consumption of saline solution. Saline solution is a sterile solution of water and salt (normally sodium chloride). The experiment involved 30 subjects, 15 who consumed one litre of water in a five minute period and 15 who consumed one litre of saline solution in the same five minute period. All subjects were required to stay in a small room maintained at a temperature of 25°C and were asked to perform minimal physical activity. Urine production over the three hours following fluid consumption was recorded for all subjects. The results for each group were averaged and are presented below.

Time (minutes)	**Urine production (mL)**	
	Water consumption	Saline solution consumption
0	24	18
30	360	21
60	450	27
90	255	36
120	48	29
150	30	34
180	27	24

(a) (i) Propose a hypothesis for the experiment. [1]

(ii) List **two** variables that were controlled in the experiment. [2]

(b) Suggest how researchers could increase the

(i) validity of the experiment. [1]

(ii) reliability of the results. [1]

(c) Graph the data in the table on the grid below. [5]

9. **[9 marks]** *(2015:2.36)*

Apnoea of prematurity is a common condition in premature infants (infants born earlier than 38 weeks gestation). Apnoea results in a complete stoppage in breathing for up to 20 seconds or more. Apnoea has various causes but if left untreated can result in low blood pressure and heart rate, brain damage, and sometimes death. Infants with apnoea of prematurity often require help with their breathing, which may include insertion of a breathing tube (intubation) or being supplied with oxygen-rich air in the humidicrib. One of the most common drug treatments for apnoea of prematurity is daily doses of caffeine, which helps to stimulate the breathing reflex and eliminate the symptoms of apnoea.

Presented below are some of the results from an experiment on the effects of caffeine in premature infants. In group A, 960 infants were administered daily doses of oral caffeine. In group B, 932 infants were administered daily doses of an oral placebo. All infants in the experiment were born between 26 and 28 weeks gestational age. All were given daily doses of either caffeine or the placebo for six weeks.

	Mean age of infants (gestational age) in weeks	
	Group A	**Group B**
Age at first dose	27.8	28.1
Age at last dose	33.8	34.1
Age at last use of breathing tube (intubation) required	29.5	30.4
Age at last use of additional oxygen required	33.7	35.3

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